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United States Patent Application Publication	20250271731
Kind Code	A1
Publication Date	August 28, 2025
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PROJECTION LENS MODULE AND PROJECTION DEVICE

Abstract

A projection lens module including a lens barrel, a lens element assembly, a focus ring and a shielding member is provided. The lens barrel has an accommodation space and a light exit opening. The lens element assembly is disposed in the accommodation space. The focus ring is disposed on an outer side of the lens barrel, and the focus ring is configured to drive the lens barrel to rotate relative to an optical axis of the lens element assembly. The shielding member is disposed on the outer side of the lens barrel. The shielding member is connected to the focus ring and covers a part of the light exit opening of the lens barrel. The shielding member is configured to move in a first direction with the focus ring, and a relative distance between the shielding member and the focus ring on the optical axis is unchanged.

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Appl. No.: **19/047640**

Filed: **February 07, 2025**

Foreign Application Priority Data

CN	202410433154.3	Apr. 11, 2024

Related U.S. Application Data

us-provisional-application US 63556845 20240222

Publication Classification

Int. Cl.: **G03B11/04** (20210101); **G02B7/02** (20210101); **G03B21/14** (20060101)

U.S. Cl.:

CPC **G03B11/045** (20130101); **G02B7/021** (20130101); **G03B21/142** (20130101); **G03B21/145** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the priority benefit of U.S. provisional application Ser. No. 63/556,845 filed on Feb. 22, 2024, and China application serial no. 202410433154.3 filed on Apr. 11, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND
Technical Field
[0002] The invention relates to an optical module and an electronic device, and particularly relates to a projection lens module and projection device.
Description of Related Art
[0003] Projection device is a display device used to produce large-size images, which has been constantly improved along with evolution and innovation of science and technology. An imaging principle of the projection device is to convert an illumination beam generated by an illumination system into an image beam through a light valve, and then project the image beam onto a projection target (such as a screen or wall) through a projection lens to form a projection image.
[0004] The projection device transmits the illumination beam to the light valve via the illumination system, the light valve forms the image beam and reflect the image beam out of the projection lens. When the image beam is reflected by the light valve, some stray light (or reflected light) may cause light and shadow in an invalid projection region outside an effective projection region for human eyes to observe. As a result, user's viewing experience when viewing images in the effective projection region may be affected. Therefore, it is necessary to use a shielding plate to block the stray light. However, an existing shielding plate design has problems of requiring manual installation or removal, increasing an overall volume of the projection device, and insufficient reliability.
[0005] The information disclosed in this Background section is only for enhancement of understanding of the background of the described technology and therefore it may contain information that does not form the prior art that is already known to a person of ordinary skill in the art. Further, the information disclosed in the Background section does not mean that one or more problems to be resolved by one or more embodiments of the invention was acknowledged by a person of ordinary skill in the art.
SUMMARY
[0006] The invention is directed to a projection lens module and a projection device, which are adapted to continuously block stray light at a specific position of a light exit opening of the projection lens module when adjusting a focal length, thereby improving optical quality of projection.
[0007] Additional aspects and advantages of the present invention will be set forth in the description of the techniques disclosed in the invention.
[0008] In order to achieve one or a portion of or all of the objects or other objects, an embodiment of the invention provides a projection lens module including a lens barrel, a lens element assembly, a focus ring and a shielding member. The lens barrel has an accommodation space and a light exit

opening. The lens element assembly is disposed in the accommodation space. The focus ring is disposed on an outer side of the lens barrel, and the focus ring is configured to drive the lens barrel to rotate relative to an optical axis of the lens element assembly. The shielding member is disposed on the outer side of the lens barrel. The shielding member is connected to the focus ring and covers a part of the light exit opening of the lens barrel. The shielding member is configured to move in a first direction with the focus ring, and a relative distance between the shielding member and the focus ring on the optical axis is unchanged.

[0009] In order to achieve one or a portion of or all of the objects or other objects, an embodiment of the invention provides a projection device including an illumination system, at least one light valve and a projection lens module. The illumination system is configured to provide an illumination beam. The at least one light valve is disposed on a transmission path of the illumination beam to convert the illumination beam into an image beam. The projection lens module is disposed on a transmission path of the image beam and is configured to project the image beam out of the projection device. The projection lens module includes a lens barrel, a lens element assembly, a focus ring and a shielding member. The lens barrel has an accommodation space and a light exit opening. The lens element assembly is disposed in the accommodation space. The focus ring is disposed on an outer side of the lens barrel, and the focus ring is configured to drive the lens barrel to rotate relative to an optical axis of the lens element assembly. The shielding member is disposed on the outer side of the lens barrel. The shielding member is connected to the focus ring and covers a part of the light exit opening of the lens barrel. The shielding member is configured to move in a first direction with the focus ring, and a relative distance between the shielding member and the focus ring on the optical axis is unchanged.

[0010] Based on the above descriptions, the embodiments of the invention have at least one of the following advantages or effects. The projection lens module and projection device of the embodiments the invention, the shielding member configured to block stray light is mechanically designed so that the shielding member and the focus ring are adapted to move in the same direction, and the relative distance between the shielding member and the focus ring on the optical axis is unchanged. Therefore, when the projection lens module adjusts the focal length, there is no need to additionally disassemble or move the shielding member. In this way, when the projection lens module adjusts the focal length, the stray light at a specific position of the light exit opening of the projection lens module is continuously blocked, thereby improving the optical quality of projection.

[0011] Other objectives, features and advantages of the present invention will be further understood from the further technological features disclosed by the embodiments of the present invention wherein there are shown and described preferred embodiments of this invention, simply by way of illustration of modes best suited to carry out the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The accompanying drawings are included to provide a further understanding of the invention, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

[0013] FIG. **1** is a schematic diagram of a projection device according to an embodiment of the invention.

[0014] FIG. **2** is a partial three-dimensional view of a projection device according to an embodiment of the invention.

[0015] FIG. **3** is a schematic three-dimensional view of a projection lens module of FIG. **2**.

[0016] FIG. **4** is a schematic front view of the projection lens module of FIG. **3**.
[0017] FIG. **5** is a schematic cross-sectional view of the projection lens module of FIG. **3**.
[0018] FIG. **6** is a partial three-dimensional exploded view of the projection device in FIG. **2**.
[0019] FIG. **7** is a schematic three-dimensional view of a projection lens module according to another embodiment of the invention.
[0020] FIG. **8**A is a schematic three-dimensional view of a partial structure of the projection lens module of FIG. **7**.
[0021] FIG. **8**B is a schematic front view of a shielding member according to an embodiment of the invention.
[0022] FIG. **9** is a schematic rear view of the projection lens module of FIG. **7**.
[0023] FIG. **10** is a schematic cross-sectional view of the projection lens module of FIG. **7**.
[0024] FIG. **11** is a schematic three-dimensional exploded view of the projection lens module of FIG. **7**.
[0025] FIG. **12** is a schematic partial front view of the projection lens module of FIG. **7**.
[0026] FIG. **13** is a schematic three-dimensional view of a partial structure of a projection lens module according to another embodiment of the invention.
[0027] FIG. **14** is a schematic partial front view of the projection lens module of FIG. **13**.

DESCRIPTION OF THE EMBODIMENTS

[0028] In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings which form a part hereof, and in which are shown by way of illustration specific embodiments in which the invention may be practiced. In this regard, directional terminology, such as “top,” “bottom,” “front,” “back,” etc., is used with reference to the orientation of the Figure(s) being described. The components of the present invention can be positioned in a number of different orientations. As such, the directional terminology is used for purposes of illustration and is in no way limiting. On the other hand, the drawings are only schematic and the sizes of components may be exaggerated for clarity. It is to be understood that other embodiments may be utilized and structural changes may be made without departing from the scope of the present invention. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless limited otherwise, the terms “connected,” “coupled,” and “mounted” and variations thereof herein are used broadly and encompass direct and indirect connections, couplings, and mountings. Similarly, the terms “facing,” “faces” and variations thereof herein are used broadly and encompass direct and indirect facing, and “adjacent to” and variations thereof herein are used broadly and encompass directly and indirectly “adjacent to”. Therefore, the description of “A” component facing “B” component herein may contain the situations that “A” component directly faces “B” component or one or more additional components are between “A” component and “B” component. Also, the description of “A” component “adjacent to” “B” component herein may contain the situations that “A” component is directly “adjacent to” “B” component or one or more additional components are between “A” component and “B” component. Accordingly, the drawings and descriptions will be regarded as illustrative in nature and not as restrictive.
[0029] FIG. **1** is a schematic diagram of a projection device according to an embodiment of the invention. Referring to FIG. **1**, the embodiment provides a projection device (projector) **10**, which includes an illumination system **50**, at least one light valve **60** and a projection lens module **100**. The illumination system **50** is configured to provide an illumination beam LB. The at least one light valve **60** is disposed on a transmission path of the illumination beam LB to convert the illumination beam LB into an image beam LI. The projection lens module **100** is disposed on a transmission path of the image beam LI, and is configured to project the image beam LI out of the projection device **10** to a projection target (not shown), such as a screen or a wall.

[0030] The illumination system **50** is configured to provide the illumination beam LB. For example, in the embodiment, the illumination system **50** includes a plurality of light-emitting elements. The illumination system **50** is configured to provide illumination beams LB with beams of different wavelengths. The plurality of light-emitting elements are, for example, light-emitting diodes (LEDs), laser diodes (LDs), or a combination thereof. The illumination system **50** may also be equipped with optical elements such as a wavelength conversion element (for example, phosphor wheel), a light uniformizing element (for example, rod, or lens array), a filter element (for example, filter wheel), a plurality of light splitting and combining (for example, dichroic mirror) elements, etc. The embodiments of the invention do not limit the type or form of the illumination system **50** in the projection device **10**, and detailed structure and implementation thereof may be provided with sufficient teachings, suggestions and implementation instructions from common knowledge in the technical field, which will not be repeated.

[0031] The light valve **60** is, for example, a reflective optical modulator such as a liquid crystal on silicon panel (LCoS panel) or a digital micro-mirror device (DMD), etc. In some embodiments, the light valve **60** may also be a transmissive optical modulator such as a transparent liquid crystal panel, an electro-optical modulator, a magneto-optic modulator, or an acousto-optic modulator, etc. The embodiments of the invention does not limit a pattern and type of the light valve **60**. Detailed steps and implementation methods of the light valve **60** converting the illumination beam LB into the image beam LI may be provided with sufficient teachings, suggestions and implementation instructions from common knowledge in the technical field, which will not be repeated. In different embodiments, a number of light valves **60** may be designed from one to three.

[0032] FIG. **2** is a partial three-dimensional view of a projection device according to an embodiment of the invention. FIG. **3** is a schematic three-dimensional view of a projection lens module of FIG. **2**. FIG. **4** is a schematic front view of the projection lens module of FIG. **3**. FIG. **5** is a schematic cross-sectional view of the projection lens module of FIG. **3**. FIG. **6** is a partial three-dimensional exploded view of the projection device in FIG. **2**. Referring to FIG. **2** to FIG. **6**, the projection lens module **100** of the embodiment may be applied to at least the projection device **10** shown in FIG. **1**. In the embodiment, the projection lens module **100** includes a lens barrel **110**, a lens element assembly **120**, a focus ring **130** and a shielding member **140**. The lens barrel **110** has an accommodation space **112** and a light exit opening **114**. The lens element assembly **120** is disposed in the accommodation space **112** of the lens barrel **110**. The lens element assembly **120** includes, for example, a combination of one or more optical lenses with diopter, such as various combinations of non-planar lenses including a biconcave lens, a biconvex lens, a concavo-convex lens, a convexo-concave lens, a plano-convex lens, a plano-concave lens, etc. In an embodiment, the lens element assembly **120** may further include a planar optical lens to project the image beam LI from the light valve **60** to a projection target in a reflective manner. The optical pattern and type of the lens element assembly **120** are not limited by the embodiment of the invention.

[0033] The focus ring **130** is disposed on an outer side of the lens barrel **110**, and is engaged with the lens barrel **110**. The focus ring **130** is configured to drive the lens barrel **110** to rotate relative to an optical axis I of the lens element assembly **120**, thereby adjusting a focal length of the projection lens module **100**. In the embodiment, the focus ring **130** includes a positioning rod **132** located on an outer side of the focus ring **130** in a circumferential direction, and the positioning rod **132** extends from the focus ring **130** in a direction away from the optical axis I. The shielding member **140** is disposed on the outer side of the lens barrel **110** and the focus ring **130**. The shielding member **140** is connected to the focus ring **130** and covers a part of the light exit opening **114** of the lens barrel **110** to block stray light at a specific position of the light exit opening **114** of the projection lens module **100** to improve optical quality of projection. The shielding member **140** is configured to move with the focus ring **130** along a first direction D**1** (a positive Z-axis direction or a negative Z-axis direction) parallel to the optical axis I, and a relative distance between the shielding member **140** and the focus ring **130** on the optical axis I is unchanged. The shielding member **140** is, for example, a mask cover. The first direction D**1**, a second direction D**2** (a positive Y-axis direction or a negative Y-axis direction) which perpendicular to the first direction D**1**, and a third direction D**3** (a positive X-axis direction or a negative X-axis direction) which perpendicular to the first direction D**1** and the second direction D**2** are further defined.

[0034] In the embodiment, the focus ring **130** is located between the lens barrel **110** and the shielding member **140**, and the shielding member **140** covers a part of the focus ring **130**. The shielding member **140** includes a shielding plate **142** and a collar portion **144** connected to the periphery of the shielding plate **142**. In the embodiment, the collar portion **144** of the shielding member **140** has a sliding groove **1442**, and the positioning rod **132** of the focus ring **130** is disposed through the sliding groove **1442** of the collar portion **144**. The sliding groove **1442** is an arc-shaped slot on a side surface of the collar portion **144** in the circumferential direction, and an extending direction of the sliding groove **1442** is perpendicular to the extending direction of the optical axis I. When the focus ring **130** of the projection lens module **100** is rotated to adjust a focal length, as the positioning rod **132** of the focus ring **130** is disposed through the sliding groove **1442** of the collar portion **144**, the positioning rod **132** is moved in the sliding groove **1442** of the collar portion **144**, and the sliding groove **1442** may limit the movement of the positioning rod **132** in the first direction D**1**, thereby keeping the relative distance between the shielding member **140** and the focus ring **130** on the optical axis I unchanged. In other words, when the projection lens module **100** adjusts the focal length, there is no need to additionally disassemble or move the shielding member **140**. In this way, when the projection lens module **100** adjusts the focal length, the stray light at the specific position of the light exit opening **114** of the projection lens module **100** may be continuously blocked, thereby improving the optical quality of projection.

[0035] In the embodiment, the shielding plate **142** of the shielding member **140** is arcuate, and the projection lens module **100** further includes a decorative ring (deco ring) **150**, the decorative ring **150** is disposed on the outer side of the lens barrel **110**. The decorative ring **150** is located between the lens barrel **110** and a housing **12** of the projection device **10** to fix a relative position of the projection lens module **100** and the housing **12**. The decorative ring **150** includes two position-limiting members **152**, and the position-limiting members **152** are, for example, plate-shaped, such as stoppers. Both ends of the collar portion **144** of the shielding member **140** respectively abut against the two position-limiting members **152**, as shown in FIG. **4**. In this way, when the focus ring **130** of the projection lens module **100** is rotated to adjust the focal length, the collar portion **144** of the shielding member **140** abuts against the two position-limiting members **152** of the decorative ring **150**, preventing the shielding member **140** from being rotated by the focusing ring **130**. As a result, the shielding member **140** is not driven by the focus ring **130** to rotate, and the shielding member **140** may not move along the positive Y-axis direction or the negative Y-axis direction with the focus ring **130**. The shielding member **140** may move back and forth (along the positive Z-axis direction or the negative Z-axis direction) together with the lens element assembly **120**, so that the region covered by the shielding plate **142** in the projection lens module **100** remains at a position below the optical axis I on a reference plane (XY plane) perpendicular to the optical axis I, and the shielding member **140**, being disposed on a transmission path of the stray light may block the stray light projected out of the projection device **10**, so that a bright region originally formed by accumulation of stray light is eliminated. In an embodiment, the above-mentioned position-limiting member **152** may not be disposed on the decorative ring **150**, but may be disposed on the housing **12** of the projection device **10**.

[0036] FIG. **7** is a schematic three-dimensional view of a projection lens module according to another embodiment of the invention. FIG. **8**A is a schematic three-dimensional view of a partial structure of the projection lens module of FIG. **7**. FIG. **9** is a schematic rear view of the projection lens module of FIG. **7**. FIG. **10** is a schematic cross-sectional view of the projection lens module of FIG. **7**. FIG. **11** is a schematic three-dimensional exploded view of the projection lens module of FIG. **7**. FIG. **12** is a schematic partial front view of the projection lens module of FIG. **7**. Referring to FIG. **7**, FIG. **8**A and FIG. **9** to FIG. **12**, a projection lens module **100**A of the embodiment is similar to the projection lens module **100** shown in FIG. **2**. A difference there between is that in the embodiment, a shielding member **140**A is located between the lens barrel **110** and the focus ring **130**. The focus ring **130** covers a part of the shielding member **140**A. In the embodiment, the shielding member **140**A includes a shielding plate **142**, a collar portion **144** connected to the periphery of the shielding plate **142**, and a positioning portion **146** connected to the collar portion **144** and protruding along the second direction D**2**. The second direction D**2** is perpendicular to the first direction D**1**. The shielding member **140**A further includes a connection portion **145** extending along the first direction D**1** between the collar portion **144** and the positioning portion **146**.

[0037] In the embodiment, the lens barrel **110** includes a plurality of first clamping posts **116**. The first clamping posts **116** are located on the outer side of the lens barrel **110** in the circumferential direction and extend in a direction away from the optical axis I. In the embodiment, a number of the plurality of first clamping posts **116** is, for example, six, and they are evenly distributed on the outer side of the lens barrel **110**. The first clamping posts **116** are, for example, posts or pins. The collar portion **144** of the shielding member **140**A has a sliding groove **1442**, and at least one of the first clamping posts **116** is disposed through the sliding groove **1442** of the collar portion **144**. In the embodiment, the first clamping post **116** disposed through the sliding groove **1442** has a locking hole **1162**, and the projection lens module **100**A further includes a locking accessory **160**, which is locked to the locking hole **1162** of the first clamping post **116** to limit the movement of the shielding member **140**A. The locking accessory **160** is, for example, a screw. The locking hole **1162** is, for example, a screw hole. When the focus ring **130** of the projection lens module **100**A is rotated to adjust the focal length, as the first clamping post **116** of the lens barrel **110** is disposed through the sliding groove **1442** of the collar portion **144**, the first clamping post **116** moves in the sliding groove **1442** of the collar portion **144**, and the relative distance between the shielding member **140**A and the focus ring **130** on the optical axis I is unchanged. The shielding member **140**A moves simultaneously with the lens barrel **110** in the direction of the optical axis I, and the blocking member **140**A may not be driven to rotate due to the rotation of the focus ring **130**. In other words, when the projection lens module **100**A adjusts the focal length, there is no need to additionally disassemble or move the shielding member **140**A. In this way, when the projection lens module **100**A adjusts the focal length, the stray light at a specific position of the light exit opening **114** of the projection lens module **100**A may be continuously blocked, thereby improving the optical quality of projection.

[0038] In the embodiment, the projection lens module **100**A further includes an adapter ring **170**, which is disposed between the lens barrel **110** and the shielding member **140**A. The adapter ring **170** includes a plurality of clamping slots **172**, and the plurality of first clamping posts **116** of the lens barrel **110** are disposed through the plurality of clamping slots **172**, and one of the plurality of first clamping posts **116** disposed through the clamping slot **172** is further disposed through the sliding groove **1442** of the collar portion **144**. The clamping slots **172** are, for example, slots. An outer surface of the adapter ring **170** in the circumferential direction includes at least one second clamping post **174**, and an inner surface of the focus ring **130** in the circumferential direction includes at least one clamping groove structure (for example, a groove) **134**. In the embodiment, a number of the second clamping posts **174** and the clamping groove structures **134** is, for example, three or six, and the second clamping posts **174** are evenly distributed on the outer side of the adapter ring **170** in the circumferential direction. The second clamping post **174** is for example, a post or a pin. The second clamping post **174** of the adapter ring **170** is engaged with the clamping groove structure **134** of the focus ring **130**. In an embodiment, the clamping groove structure **134** includes a stop portion (not shown in FIG) and an inclined portion (not shown in FIG) connected to each other. The inclined portion has a design that tapers along with a distance relative to the adapter ring **170**. The inclined portion includes a closest end that is closest to the adapter ring **170** and a farthest end that is farthest from the adapter ring **170**. The stop portion is connected to the farthest end of the inclined portion. By rotating (for example, counterclockwise) the focus ring **130** or the adapter ring **170**, the second clamping post **174** of the adapter ring **170** is gradually engaged and fixed with the stop portion of the clamping groove structure **134** at the inclined portion of the clamping groove structure **134**, thereby fixing the adapter ring **170** and the focusing ring **130** to each other. In this way, the focus ring **130** may be fixed to the outer side of the lens barrel **110** through the configuration of the adapter ring **170**, and at the same time, the relative distance between the shielding member **140**A and the focus ring **130** on the optical axis I is unchanged when the projection lens module **100**A adjusts the focal length, and the shielding member **140**A is kept not rotating along with the focus ring **130** and continues to block stray light at a specific position of the light exit opening **114** of the projection lens module **100**A. In the embodiment, since the shielding member **140**A is located between the lens barrel **110** and the focus ring **130**, a circumferential side of the focus ring **130** may be used as a position for a user to grasp and rotate the focus ring **130**. In an embodiment, the projection lens module **100**A may further include a dust cover (not shown in FIG) located on the outer side of the focus ring **130**, and the focus ring **130** is located between the shielding member **140**A and the dust cover.

[0039] In the embodiment, the positioning portion **146** of the shielding member **140**A has, for example, two columnar structures extending in the direction of the housing **12** (the negative Y-axis direction), which may be clamped and fixed to the housing **12** of the projection device **10**, so that the shielding member **140**A and the focus ring **130** may move along the second direction D**2**, and the relative distance between the shielding member **140**A and the focus ring **130** in the second direction D**2** is unchanged. In the embodiment, the housing **12** of the projection device **10** includes a position-limiting structure **14**. The position-limiting structure **14** extends from the housing **12** toward the direction of the projection lens module **100**A (the positive Y-axis direction), and the positioning portion **146** of the shielding member **140**A is engaged with the position-limiting structure **14** of the housing **12**. In other words, the positioning portion **146** of the shielding member **140**A and the position-limiting structure **14** of the housing **12** have corresponding structures with opposite extending directions. The position-limiting structure **14** further has an extension structure extending parallel to the optical axis I. Therefore, in the embodiment, the position-limiting structure **14** may be configured to allow the projection lens module **100**A to move along the first direction D**1** and/or the second direction D**2**, thereby driving the shielding member **140**A to move along the first direction D**1** and/or the second direction D**2**. In this way, the shielding member **140**A of the embodiment has more dimensions of adjustment space, and the shielding member **140**A may move back and forth and/or up and down along with the lens element assembly **120**. In different embodiments, the position-limiting structure **14** may be selectively connected to or integrally formed with the body of the housing **12**.

[0040] FIG. **8**B is a schematic front view of a shielding member according to an embodiment of the invention. Referring to FIG. **8**A and FIG. **8**B, in an embodiment, the stray light is only distributed in a lower region outside an effective projection region, so that the shielding plate **142** may be correspondingly designed to block a lower side of the light exit opening **114** of the lens barrel **110**, and an outline of the shielding plate **142** is an arcuate shape. In another option of the embodiment, the outline of the shielding plate **142** of the shielding member **140**B may be modified from the arcuate shape to a design with a rectangular opening. In this way, it may be applied to situations where stray light is distributed in the lower region, upper region, left region and right region outside the effective projection region. The shielding member **140**B may be designed to expose only partial area of the light exit opening **114** of the lens barrel **110** corresponding to the effective projection region, and block the rest area of the light exit opening **114** corresponding to the effective projection region. The effective projection region refers to a region where the image beam LI projected from the projection device **10** forms an image.

[0041] FIG. **13** is a schematic three-dimensional view of a partial structure of a projection lens module according to another embodiment of the invention. FIG. **14** is a schematic partial front

view of the projection lens module of FIG. **13**. Referring to FIG. **13** and FIG. **14**, a projection lens module **100**B of the embodiment is similar to the projection lens module **100**A shown in FIG.
[0042] **8**A, and a difference there between is that in the embodiment, a shielding member **140**C further includes an elastic member **148**, and the elastic member **148** may be a compressible elastic piece made of metal (for example, a stainless steel elastic piece). The elastic member **148** is disposed on an inner side of the positioning portion **146**, and is located between the position-limiting structure **14** of the housing **12** and the positioning portion **146**, and clamps the positioning-limiting structure **14**. A width between the two columnar structures of the positioning portion **146** extending toward the housing **12** in the third direction D**3** is greater than a width of the position-limiting structure **14** of the housing **12** in the third direction D**3**. Through an elastic force of the elastic member **148**, the elastic member **148** is configured to allow the shielding member **140**C and the focus ring **130** to move along the third direction D**3**, and the relative distance between the shielding member **140**C and the focus ring **130** in the third direction D**3** is unchanged. The third direction D**3** is perpendicular to the second direction D**2** and the first direction D**1** respectively. In this way, the shielding member **140**C of the embodiment has more dimensions of adjustment space. The shielding member **140**C may move back and forth (along the first direction D**1**) and/or up and down (along the second direction D**2**) and/or move to the left or right (along the third direction D) along with the lens element assembly **120**.
[0043] In summary, the projection lens module and the projection device of the embodiments of the invention have at least one of the following advantages or effects. The projection lens module and projection device of the embodiments the invention, the shielding member configured to block stray light is mechanically designed so that the shielding member and the focus ring are adapted to move in the same direction, and the relative distance between the shielding member and the focus ring on the optical axis is unchanged. Therefore, when the projection lens module adjusts the focal length, there is no need to additionally disassemble or move the shielding member. In this way, when the projection lens module adjusts the focal length, the stray light at a specific position of the light exit opening of the projection lens module is continuously blocked, thereby improving the optical quality of projection.
[0044] The foregoing description of the preferred embodiments of the invention has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form or to exemplary embodiments disclosed. Accordingly, the foregoing description should be regarded as illustrative rather than restrictive. Obviously, many modifications and variations will be apparent to practitioners skilled in this art. The embodiments are chosen and described in order to best explain the principles of the invention and its best mode practical application, thereby to enable persons skilled in the art to understand the invention for various embodiments and with various modifications as are suited to the particular use or implementation contemplated. It is intended that the scope of the invention be defined by the claims appended hereto and their equivalents in which all terms are meant in their broadest reasonable sense unless otherwise indicated. Therefore, the term “the invention”, “the present invention” or the like does not necessarily limit the claim scope to a specific embodiment, and the reference to particularly preferred exemplary embodiments of the invention does not imply a limitation on the invention, and no such limitation is to be inferred. The invention is limited only by the spirit and scope of the appended claims. Moreover, these claims may refer to use “first”, “second”, etc. following with noun or element. Such terms should be understood as a nomenclature and should not be construed as giving the limitation on the number of the elements modified by such nomenclature unless specific number has been given. The abstract of the disclosure is provided to comply with the rules requiring an abstract, which will allow a searcher to quickly ascertain the subject matter of the technical disclosure of any patent issued from this disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Any advantages and benefits described may not apply to all embodiments of the invention. It should be

appreciated that variations may be made in the embodiments described by persons skilled in the art without departing from the scope of the present invention as defined by the following claims. Moreover, no element and component in the present disclosure is intended to be dedicated to the public regardless of whether the element or component is explicitly recited in the following claims.

Claims

1. A projection lens module, comprising a lens barrel, a lens element assembly, a focus ring and a shielding member, wherein the lens barrel has an accommodation space and a light exit opening; the lens element assembly is disposed in the accommodation space; the focus ring is disposed on an outer side of the lens barrel, and the focus ring is configured to drive the lens barrel to rotate relative to an optical axis of the lens element assembly; and the shielding member is disposed on the outer side of the lens barrel, the shielding member is connected to the focus ring and covers a part of the light exit opening of the lens barrel, the shielding member is configured to move along a first direction with the focus ring, and a relative distance between the shielding member and the focus ring on the optical axis is unchanged.
2. The projection lens module as claimed in claim 1, wherein the focus ring is located between the lens barrel and the shielding member, and the shielding member covers a part of the focus ring.
3. The projection lens module as claimed in claim 2, wherein the shielding member comprises a shielding plate and a collar portion connected to a periphery of the shielding plate.
4. The projection lens module as claimed in claim 3, wherein the focus ring comprises a positioning rod, the positioning rod is located on an outer side of the focusing ring, the collar portion of the shielding member has a sliding groove, and the positioning rod of the focus ring is disposed through the sliding groove of the collar portion.
5. The projection lens module as claimed in claim 3, wherein the shielding plate is arcuate, and the projection lens module further comprises a decorative ring disposed on the outer side of the lens barrel, the decorative ring comprises two position-limiting members, and two ends of the collar portion respectively abut against the two position-limiting members.
6. The projection lens module as claimed in claim 1, wherein the shielding member is located between the lens barrel and the focus ring, and the focus ring covers a part of the shielding member.
7. The projection lens module as claimed in claim 6, wherein the shielding member comprises a shielding plate, a collar portion connected to a periphery of the shielding plate, and a positioning portion connected to the collar portion and protruding in a second direction, the positioning portion is configured to allow the shielding member and the focus ring to move along the second direction, and a relative distance between the shielding member and the focus ring in the second direction is unchanged, wherein the second direction is perpendicular to the first direction.
8. The projection lens module as claimed in claim 7, wherein the shielding member further comprises an elastic member disposed on an inner side the positioning portion, the elastic member is configured to allow the shielding member and the focus ring to move along a third direction, and a relative distance between the shielding member and the focus ring in the third direction is unchanged, and the third direction is perpendicular to the second direction and the first direction respectively.
9. The projection lens module as claimed in claim 7, wherein the lens barrel comprises a plurality of first clamping posts located on the outer side of the lens barrel, the collar portion of the shielding member has a sliding groove, and at least one of the clamping posts is disposed through the sliding groove of the collar portion.
10. The projection lens module as claimed in claim 9, wherein at least one of the plurality of first clamping posts has a locking hole, and the projection lens module further comprises a locking accessory that is locked to the locking hole to limit movement of the shielding member.
11. The projection lens module as claimed in claim 9, wherein the projection lens module further

comprises an adapter ring disposed between the lens barrel and the shielding member, the adapter ring comprises a plurality of clamping grooves, and the plurality of first clamping posts of the lens barrel are disposed through the plurality of clamping grooves.
12. The projection lens module as claimed in claim 11, wherein an outer surface of the adapter ring comprises at least one second clamping post, an inner surface of the focus ring comprises at least one clamping groove structure, and the at least one second clamping post of the adapter ring is engaged with the at least one clamping groove structure to fix the focus ring.
13. The projection lens module as claimed in claim 7, wherein the shielding plate is arcuate or has a rectangular opening.
14. A projection device, comprising an illumination system, at least one light valve and a projection lens module, wherein: the illumination system is configured to provide an illumination beam; the at least one light valve is disposed on a transmission path of the illumination beam to convert the illumination beam into an image beam; and the projection lens module is disposed on a transmission path of the image beam and is configured to project the image beam out of the projection device, the projection lens module comprises a lens barrel, a lens element assembly, a focus ring and a shielding member, wherein the lens barrel has an accommodation space and a light exit opening; the lens element assembly is disposed in the accommodation space; the focus ring is disposed on an outer side of the lens barrel, and the focus ring is configured to drive the lens barrel to rotate relative to an optical axis of the lens element assembly; and the shielding member is disposed on the outer side of the lens barrel, the shielding member is connected to the focus ring and covers a part of the light exit opening of the lens barrel, the shielding member is configured to move along a first direction with the focus ring, and a relative distance between the shielding member and the focus ring on the optical axis is unchanged.
15. The projection device as claimed in claim 14, wherein the focus ring is located between the lens barrel and the shielding member, and the shielding member covers a part of the focus ring.
16. The projection device as claimed in claim 15, wherein the shielding member comprises a shielding plate and a collar portion connected to a periphery of the shielding plate.
17. The projection device as claimed in claim 16, wherein the focus ring comprises a positioning rod, the positioning rod is located on an outer side of the focusing ring, the collar portion of the shielding member has a sliding groove, and the positioning rod of the focus ring is disposed through the sliding groove of the collar portion.
18. The projection device as claimed in claim 16, wherein the shielding plate is arcuate, and the projection lens module further comprises a decorative ring disposed on the outer side of the lens barrel and connected to a housing of the projection device, the decorative ring comprises two position-limiting members, and two ends of the collar portion respectively abut against the two position-limiting members.
19. The projection device as claimed in claim 14, wherein the shielding member is located between the lens barrel and the focus ring, and the focus ring covers a part of the shielding member.
20. The projection device as claimed in claim 19, wherein the shielding member comprises a shielding plate, a collar portion connected to a periphery of the shielding plate, and a positioning portion connected to the collar portion and protruding in a second direction, the positioning portion is engaged with a position-limiting structure of a housing of the projection device, and the positioning portion is configured to allow the shielding member and the focus ring to move along the second direction, and a relative distance between the shielding member and the focus ring in the second direction is unchanged, wherein the second direction is perpendicular to the first direction.
21. The projection device as claimed in claim 20, wherein the position-limiting structure is configured to allow the shielding member to move along in the first direction and/or the second direction.
22. The projection device as claimed in claim 20, wherein the shielding member further comprises an elastic member disposed on an inner side the positioning portion, and located between the

position-limiting structure of the housing and the positioning portion, the elastic member is configured to move the shielding member and the focus ring in a third direction, and a relative distance between the shielding member and the focus ring in the third direction is unchanged, and the third direction is perpendicular to the second direction and the first direction respectively.

23. The projection device as claimed in claim 20, wherein the lens barrel further comprises a plurality of first clamping posts located on the outer side of the lens barrel, the collar portion of the shielding member has a sliding groove, and at least one of the clamping posts is disposed through the sliding groove of the collar portion.

24. The projection device as claimed in claim 23, wherein at least one of the plurality of first clamping posts has a locking hole, and the projection lens module further comprises a locking accessory that is locked to the locking hole to limit movement of the shielding member.

25. The projection device as claimed in claim 23, wherein the projection lens module further comprises an adapter ring disposed between the lens barrel and the shielding member, the adapter ring comprises a plurality of clamping grooves, and the plurality of first clamping posts of the lens barrel are disposed through the plurality of clamping grooves.

26. The projection device as claimed in claim 25, wherein an outer surface of the adapter ring comprises at least one second clamping post, an inner surface of the focus ring comprises at least one clamping groove structure, and the at least one second clamping post of the adapter ring is engaged with the at least one clamping groove structure to fix the focus ring.

27. The projection device as claimed in claim 20. wherein the shielding plate is arcuate or has a rectangular opening.